

RENE BENJAMIN ULLOA

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EDUCATION

Master of Arts | *Mathematics (Mathematics of Finance Specialization)*

September 2025 – Expected December 2026

Columbia University

New York, NY

• **Relevant Graduate Courses** – Stochastic Processes, Non-Linear Options Pricing, Statistical Inference, Numerical Methods

Master of Arts | *Economics*

August 2020 – December 2021

Duke University

Durham, NC

• **Relevant Graduate Courses** – Measure & Integration, Smooth Manifolds, Mathematical Finance, Continuous Time Methods, Applied

Econometrics in Microeconomics, Functional Analysis, Ordinary & Partial Differential Equations

• **Paper** – *Migration Movements Since 2000 & Its Influences*, with Xiaolin Ding

Bachelor of Science | *Mathematics & Economics*

August 2015 – May 2020

George Mason University

Fairfax, VA

• **Relevant Courses** – Real Analysis I-II, Ordinary Differential Equations (Ph.D.), Partial Differential Equations (Ph.D.), Topology of Metric

Spaces (Ph.D.), Advanced Linear Algebra, Macroeconomics, Microeconomics, Game Theory, Econometrics, Mathematical Statistics

PROFESSIONAL EXPERIENCE

Fixed Income Data Scientist (Vice President)

August 2025 – April 2026

Tradeweb Markets LLC

New York, NY

- Architected a model-agnostic, agentic LLM chatbot (Docker, Linux) with custom hooks/skills, MCP integrations, and a retrieval-augmented knowledge base over Snowflake and Kafka, delivering responses and charts via Outlook SMTP to C-suite, clients, and developers.
- Engineered MCP (Model Context Protocol) integrations for automated report delivery with dynamic charts, and built a speech-to-text agent for the sales desk to accelerate dealer outreach on RFQ+ utilization and client rankings.
- Built and operated the fixed income data platform powering cover-policy models and dealer stream monitoring across MBS, MBSP, USDIRS, CDXIG, and Treasury markets, tracking spread tightness, ladder sizes, and platform comparisons; orchestrated with Prefect and GitLab CI/CD, with Tableau Server extraction via tabcmd and Python.
- Designed and published a cross-product liquidity index tracking 1-minute transaction cost standard deviations across UST, USDIRS, MBS FNCL, and CDXIG, with composite width analytics and macro-event overlays for market structure research.
- Lead weekly AI/LLM enablement seminars on Claude Code and OpenAI Codex, driving adoption of agentic coding tools across the sales and engineering organizations.

Data & Insights Consultant (Data Science)

June 2024 – August 2025

Visa

Miami, FL

- Engineered Airflow-orchestrated ETL pipelines on Hadoop, Hive, and Spark, integrating cross-border, credit, and debit transaction sources to support analytics and machine learning initiatives for external clients and internal stakeholders.
- Built and maintained production Tableau dashboards monitoring Visa network performance, with automated refresh schedules and data quality checks across cross-border, credit, and debit transaction flows.
- Productionized statistical metrics, machine learning anomaly detection models, and an automated email alerting system surfacing authorization-rate and transaction-pattern anomalies across domestic and international markets.

Research Analyst - TMT, Semiconductors, & Industrials

August 2022 - June 2024

Point72 Asset Management, L.P.

New York, NY

- Built Airflow-orchestrated ETL and modeling pipelines on credit card, satellite imagery, email receipt, job postings, geolocation, trade, and web scrape data, expanding the scope of actionable insights for PMs across Long/Short Equity and other investment strategies.
- Sourced, evaluated, and onboarded new alternative datasets across consumer and macro-sensitive domains; worked directly with vendors to validate data quality, shape delivery formats, and stand up automated ingestion pipelines for PM use.
- Engineered predictive models using Stochastic Gradient Boosting, AdaBoost, and other ensemble methods, materially improving KPI forecast accuracy across TMT, Semiconductors, and Industrials coverage.
- Designed portfolio strategy analyses using backtesting frameworks, identifying critical investment performance drivers including factor returns, risk exposure, and turnover.

Equities Research Associate - Semiconductors, Hardtech, & Consumer Electronics

December 2021 - July 2022

M Science

New York, NY

- Built and operated ETL pipelines in PySpark and SQL on Databricks, transforming email receipt, credit card, import/export trade, web scrape, and SDK mobile app data into modeling-ready datasets feeding KPI prediction APIs for TMT and Semiconductor coverage.
- Developed statistical models and REST API endpoints serving model outputs to research and client-facing systems; published equity research reports synthesizing model-driven insights for institutional clients.

SKILLS

Programming: Python (Pandas, NumPy, SciPy, Seaborn, Sklearn, TensorFlow), PySpark, TypeScript, SQL, R, MATLAB, C++

Data Engineering & Cloud: ETL/ELT pipelines (batch and streaming), data lake / lakehouse patterns, orchestration (Airflow, Prefect), CI/CD (GitLab, GitHub Actions), testing (pytest), schema design, REST APIs; Hive, Hadoop, Spark, Databricks, Delta Lake, Snowflake, Parquet, Kafka, AWS (S3, EC2, Glue), Docker, Kubernetes, Linux, Bloomberg

AI & LLM: Claude Code, Claude Agent SDK, MCP (Model Context Protocol), OpenAI Codex, RAG, agentic systems (custom hooks/skills/tools), model-agnostic chatbots, speech-to-text, prompt engineering

Statistical & ML: Time-series and panel data analysis, gradient boosting (e.g., XGBoost), random forests, logistic regression, anomaly detection, feature engineering, backtesting

Visualization: Tableau, Power BI, Excel, $\LaTeX$

Languages: English (Native), Spanish (Native), Portuguese (Native), Italian (Intermediate), French (Basic)

Interests: Earnings Season, MMA, Judo, Freestyle Wrestling, Olympic Weightlifting, Golf, Neovim, Cursor, Semiconductors